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ТОМСКИЙ ГОСУДАРСТВЕННЫЙ УНИВЕРСИТЕТ (НИ ТГУ)
Институт прикладной математики и информатики

ОТЧЕТ ПО ЛАБОРАТОРНОЙ РАБОТЕ
по дисциплине «Системы управления базами данных»

Рекурсивные запросы

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1. Следующий запрос ищет случаи, когда manager_id равен NULL, возвращая корневой узел.

SELECT employee_id, last_name, department_id, manager_id, salary, hire_date FROM HR.employees WHERE manager_id IS NULL;

Worksheet | Query Builder

```
SELECT employee_id, last_name, department_id, manager_id, salary, hire_date
FROM HR.employees WHERE manager_id IS NULL;
```

Query Result

SQL | All Rows Fetched: 1 in 0,003 seconds

	EMPLOYEE_ID	LAST_NAME	DEPARTMENT_ID	MANAGER_ID	SALARY	HIRE_DATE
1	100	King	90	(null)	24000	17.06.03

2. Следующий запрос возвращает связь узлов с их непосредственными родителями.

SELECT E.last_name "Employee", M.last_name "Manager" FROM HR.employees E, HR.employees M WHERE E.manager_id = M.employee_id;

Worksheet | Query Builder

```
SELECT E.last_name "Employee", M.last_name "Manager"
FROM HR.employees E, HR.employees M WHERE E.manager_id = M.employee_id;
```

Query Result

SQL | All Rows Fetched: 106 in 0,004 seconds

	Employee	Manager
1	Bates	Cambrault
2	Bloom	Cambrault
3	Fox	Cambrault
4	Kumar	Cambrault
5	Ozer	Cambrault
6	Smith	Cambrault
7	Hunold	De Haan

3. Вывести всех сотрудников и их менеджеров.

SELECT E.last_name "Employee", M.last_name "Manager" FROM HR.employees E LEFT JOIN HR.employees M ON E.manager_id = M.employee_id;

Worksheet | Query Builder

```
SELECT E.last_name "Employee", M.last_name "Manager"
FROM HR.employees E LEFT JOIN HR.employees M ON E.manager_id = M.employee_id;
```

Query Result

SQL | All Rows Fetched: 107 in 0,006 seconds

	Employee	Manager
1	Ozer	Cambrault
2	Bloom	Cambrault
3	Fox	Cambrault
4	Smith	Cambrault
5	Bates	Cambrault
6	Kumar	Cambrault
7	Hunold	De Haan

4. Вывести все записи о сотрудниках, у которых нет подчиненных.

SELECT * FROM HR.employees WHERE employee_id NOT IN (SELECT manager_id FROM HR.employees);

Worksheet | Query Builder

```
SELECT * FROM HR.employees
WHERE employee_id NOT IN (SELECT manager_id FROM HR.employees);
```

Query Result

SQL | All Rows Fetched: 0 in 0,001 seconds

EMPLOYE...	FIRST_NA...	LAST_NAME	EMAIL	PHONE_N...	HIRE_DATE	JOB_ID	SALARY	COMMISS...	MANAGER...	DEPARTM...

SELECT employee_id, last_name, department_id, manager_id, salary, hire_date FROM HR.employees WHERE employee_id NOT IN (SELECT manager_id FROM HR.employees WHERE manager_id IS NOT NULL);

Worksheet | Query Builder

```
SELECT employee_id, last_name, department_id, manager_id, salary, hire_date
FROM HR.employees WHERE employee_id
NOT IN (SELECT manager_id FROM HR.employees WHERE manager_id IS NOT NULL);
```

Query Result

SQL | All Rows Fetched: 89 in 0,005 seconds

	EMPLOYEE_ID	LAST_NAME	DEPARTMENT_ID	MANAGER_ID	SALARY	HIRE_DATE
1	198	OConnell	50	124	2600	21.06.07
2	199	Grant	50	124	2600	13.01.08
3	200	Whalen	10	101	4400	17.09.03
4	202	Fay	20	201	6000	17.08.05
5	203	Mavris	40	101	6500	07.06.02
6	204	Baer	70	101	10000	07.06.02
7	206	Gietz	110	205	8300	07.06.02

SELECT employee_id, last_name, department_id, manager_id, salary, hire_date FROM HR.employees E WHERE NOT EXISTS (SELECT employee_id FROM HR.employees E1 WHERE E.employee_id = E1.manager_id);

Worksheet | Query Builder

```
SELECT employee_id, last_name, department_id, manager_id, salary, hire_date
FROM HR.employees E WHERE NOT EXISTS
(SELECT employee_id FROM HR.employees E1 WHERE E.employee_id = E1.manager_id);
```

Query Result

SQL | All Rows Fetched: 89 in 0,003 seconds

	EMPLOYEE_ID	LAST_NAME	DEPARTMENT_ID	MANAGER_ID	SALARY	HIRE_DATE
1	198	OConnell	50	124	2600	21.06.07
2	199	Grant	50	124	2600	13.01.08
3	200	Whalen	10	101	4400	17.09.03
4	202	Fay	20	201	6000	17.08.05
5	203	Mavris	40	101	6500	07.06.02
6	204	Baer	70	101	10000	07.06.02
7	206	Gietz	110	205	8300	07.06.02

5. Найти каждого сотрудника вместе с его менеджером.

SELECT e_top.last_name, e_2.last_name, e_3.last_name, e_4.last_name FROM HR.employees e_top LEFT OUTER JOIN HR.employees e_2 ON e_top.employee_id = e_2.manager_id LEFT OUTER JOIN HR.employees e_3 ON e_2.employee_id = e_3.manager_id LEFT OUTER JOIN HR.employees e_4 ON e_3.employee_id = e_4.manager_id WHERE e_top.manager_id IS NULL;

Worksheet | Query Builder

```
SELECT e_top.last_name, e_2.last_name, e_3.last_name, e_4.last_name
FROM HR.employees e_top
LEFT OUTER JOIN HR.employees e_2 ON e_top.employee_id = e_2.manager_id
LEFT OUTER JOIN HR.employees e_3 ON e_2.employee_id = e_3.manager_id
LEFT OUTER JOIN HR.employees e_4 ON e_3.employee_id = e_4.manager_id
WHERE e_top.manager_id IS NULL;
```

Query Result

All Rows Fetched: 89 in 0,005 seconds

	LAST_NAME	LAST_NAME_1	LAST_NAME_2	LAST_NAME_3
1	King	Hartstein	Fay	(null)
2	King	Kochhar	Whalen	(null)
3	King	Kochhar	Mavris	(null)
4	King	Kochhar	Baer	(null)
5	King	Kochhar	Higgins	Gietz
6	King	Kochhar	Greenberg	Faviet

SELECT last_name, employee_id, manager_id FROM HR.employees START WITH manager_id IS NULL CONNECT BY PRIOR employee_id = manager_id;

Worksheet | Query Builder

```
SELECT last_name, employee_id, manager_id FROM HR.employees
START WITH manager_id IS NULL CONNECT BY PRIOR employee_id = manager_id;
```

Query Result

All Rows Fetched: 107 in 0,006 seconds

	LAST_NAME	EMPLOYEE_ID	MANAGER_ID
1	King	100	(null)
2	Kochhar	101	100
3	Greenberg	108	101
4	Faviet	109	108
5	Chen	110	108
6	Sciarra	111	108
7	Urman	112	108
8	Popp	113	108

SELECT last_name, employee_id, manager_id FROM HR.employees START WITH manager_id IS NULL CONNECT BY manager_id = PRIOR employee_id;

Worksheet | Query Builder

```
SELECT last_name, employee_id, manager_id FROM HR.employees
START WITH manager_id IS NULL CONNECT BY manager_id = PRIOR employee_id;
```

Query Result

All Rows Fetched: 107 in 0,006 seconds

	LAST_NAME	EMPLOYEE_ID	MANAGER_ID
1	King	100	(null)
2	Kochhar	101	100
3	Greenberg	108	101
4	Faviet	109	108
5	Chen	110	108
6	Sciarra	111	108
7	Urman	112	108
8	Popp	113	108

6. Найти всех сотрудников и его менеджером, начиная с сотрудников с фамилией Jones.

SELECT last_name, employee_id, manager_id FROM HR.employees START WITH last_name = 'Jones' CONNECT BY manager_id = PRIOR employee_id;

Worksheet | Query Builder

```
SELECT last_name, employee_id, manager_id FROM HR.employees
START WITH last_name = 'Jones' CONNECT BY manager_id = PRIOR employee_id;
```

Query Result

All Rows Fetched: 1 in 0,004 seconds

	LAST_NAME	EMPLOYEE_ID	MANAGER_ID
1	Jones	195	123

7. Найти всех сотрудников и его менеджером, начиная с тех, кто устроился в компанию раньше всех.

SELECT last_name, employee_id, manager_id FROM HR.employees START WITH hire_date = (SELECT MIN(hire_date) FROM HR.employees) CONNECT BY manager_id = PRIOR employee_id;

Worksheet | Query Builder

```
SELECT last_name, employee_id, manager_id FROM HR.employees
START WITH hire_date = (SELECT MIN(hire_date) FROM HR.employees)
CONNECT BY manager_id = PRIOR employee_id;
```

Query Result

All Rows Fetched: 6 in 0,004 seconds

	LAST_NAME	EMPLOYEE_ID	MANAGER_ID
1	De Haan	102	100
2	Hunold	103	102
3	Ernst	104	103
4	Austin	105	103
5	Pataballa	106	103
6	Lorentz	107	103

8. Найти каждого сотрудника вместе с его менеджером, указав уровень иерархии.

SELECT level,last_name, employee_id, manager_id FROM HR.employees START WITH manager_id IS NULL CONNECT BY manager_id = PRIOR employee_id;

Worksheet | Query Builder

```
SELECT level,last_name, employee_id, manager_id FROM HR.employees
START WITH manager_id IS NULL CONNECT BY manager_id = PRIOR employee_id;
```

Query Result

All Rows Fetched: 107 in 0,005 seconds

	LEVEL	LAST_NAME	EMPLOYEE_ID	MANAGER_ID
1	1	King	100	(null)
2	2	Kochhar	101	100
3	3	Greenberg	108	101
4	4	Faviet	109	108
5	4	Chen	110	108
6	4	Sciarra	111	108
7	4	Urman	112	108

9. Найти разницу зарплаты менеджера и его подчиненного.

SELECT E.last_name, E.employee_id, E.manager_id, (SELECT M.salary - E.salary FROM HR.employees M WHERE M.employee_id = E.manager_id) FROM HR.employees E START WITH manager_id IS NULL CONNECT BY manager_id = PRIOR employee_id;

Worksheet | Query Builder

```
SELECT E.last_name, E.employee_id, E.manager_id,
(SELECT M.salary - E.salary FROM HR.employees M WHERE M.employee_id = E.manager_id)
FROM HR.employees E
START WITH manager_id IS NULL CONNECT BY manager_id = PRIOR employee_id;
```

Query Result | SQL | All Rows Fetched: 107 in 0,006 seconds

	LAST_NAME	EMPLOYEE_ID	MANAGER_ID	(SELECTM.SALARY-E.SALARYFROMHR.EMPLOYEESMWHEREM.EMPLOYEE_ID=E.MANAGER_ID)
1	King	100	(null)	(null)
2	Kochhar	101	100	7000
3	Greenberg	108	101	4992
4	Faviet	109	108	3008
5	Chen	110	108	3808

10. Вывести сотрудников в соответствии иерархии должностей.

SELECT level,last_name, first_name FROM HR.employees START WITH manager_id IS NULL CONNECT BY manager_id = PRIOR employee_id ORDER BY level;

Worksheet | Query Builder

```
SELECT level,last_name, first_name FROM HR.employees
START WITH manager_id IS NULL CONNECT BY manager_id = PRIOR employee_id
ORDER BY level;
```

Query Result | SQL | All Rows Fetched: 107 in 0,005 seconds

	LEVEL	LAST_NAME	FIRST_NAME
1	1	King	Steven
2	2	De Haan	Lex
3	2	Raphaely	Den
4	2	Weiss	Matthew
5	2	Fripp	Adam
6	2	Kaufling	Payam

11. Найти каждого сотрудника вместе с его менеджером, которые работают в отделах 10, 20 или 30.

SELECT level, last_name, employee_id, manager_id FROM HR.employees WHERE department_id IN (10,20,30) START WITH manager_id IS NULL CONNECT BY manager_id = PRIOR employee_id ORDER BY level;

Worksheet | Query Builder

```
SELECT level, last_name, employee_id, manager_id FROM HR.employees
WHERE department_id IN (10,20,30)
START WITH manager_id IS NULL CONNECT BY manager_id = PRIOR employee_id
ORDER BY level;
```

Query Result | SQL | All Rows Fetched: 9 in 0,004 seconds

	LEVEL	LAST_NAME	EMPLOYEE_ID	MANAGER_ID
1	2	Hartstein	201	100
2	2	Raphaely	114	100
3	3	Baida	116	114
4	3	Fay	202	201
5	3	Himuro	118	114
6	3	Colmenares	119	114
7	3	Khoo	115	114
8	3	Whalen	200	101
9	3	Tobias	117	114